Shaukat Aziz
Sr no: 22171 — BSc (Research) in Physics

Tera-QuaNTA Lab, IAP, IISc Bangalore
Thesis Instructor: Prof. Manukumara Manjappa

2026

Photonic Crystals & Band Theory

Maxwell Eigenvalue Problem

In an isotropic, non-magnetic medium:

$$-i\nabla \times \mathbf{E} = \omega\mu_0\mathbf{H}, \quad i\nabla \times \mathbf{H} = \omega\varepsilon\mathbf{E}$$

For a 2D slab, the modes **decouple** into:

- **TM**: nonzero (H_x, H_y, E_z)
- **TE**: nonzero (E_x, E_y, H_z)

TE Mode Equation (ours)

$$\omega^2\mu_0 H_z = -\nabla_\perp \cdot \left(\frac{1}{\varepsilon}\nabla_\perp H_z\right)$$

Solved self-consistently for the periodic dielectric $\varepsilon(\mathbf{r})$. By Bloch's theorem: $H_z(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}}u_{n,\mathbf{k}}(\mathbf{r})$

Why TE (Hz) not TM?

1. TE modes yield a **wider** and more robust photonic band gap in dielectric-hole structures
2. H_z antinodes localise at **unit-cell corners** at the M-point of BZ — exactly matching the SSH atom sites
3. TE slab modes have **lower out-of-plane radiation loss** (higher Q) than TM
4. HOTI topological invariant (quadrupole moment q_{xy}) is defined for the **TE band manifold**

Multipole Expansion of Slab Modes

Cylindrical Multipole Expansion (Supplementary Note 1)

For a structure with C_n point group symmetry, H_z is expanded as:

$$H_z(\mathbf{r}_\perp) = \sum_m f_m(r)\, e^{im\varphi}$$

Rotational invariance under $\varphi \to \varphi + 2\pi/n$ requires:

$$H_z(r, \varphi + 2\pi/n) = H_z(r, \varphi)\, e^{i2\pi l/n}$$

where l is the **angular momentum quantum number**.

Allowed Multipole Coefficients

Only $m = l + qn$ (for integer q) contribute to modes with angular momentum l. For C_5: $l \in \{0, \pm 1, \pm 2\}$, giving **5 in-gap states**.

Electric Field from H_z

$$\mathbf{E} = \frac{1}{r}\partial_\varphi H_z\, \hat{r} - \partial_r H_z\, \hat{\varphi}$$

$l = 0$: polarisation vortex (TM-like electric field)
$l = \pm 1$: vector vortex (OAM $q = \pm 1$)
$l = \pm 2$: higher-OAM beam

1D Su-Schrieffer-Heeger (SSH) Model

Bloch Hamiltonian

Two sublattices A, B with alternating hoppings t_1 (intra-cell) and t_2 (inter-cell):

$$H(k) = (t_1 + t_2 \cos k)\,\sigma_x + t_2 \sin k\,\sigma_y$$

Eigenvalues:

$$E_\pm(k) = \pm\sqrt{t_1^2 + t_2^2 + 2t_1 t_2 \cos k}$$

Topological Invariant: Zak Phase

$$\gamma = i \oint_{\mathrm{BZ}} \langle u_k | \partial_k | u_k \rangle \, dk = \begin{cases} 0 & \text{trivial } (|t_1| > |t_2|) \\ \pi & \text{topological } (|t_2| > |t_1|) \end{cases}$$

Winding Number

$$w = \frac{1}{2\pi i} \oint \frac{d}{dk} \ln\left(t_1 + t_2 e^{ik}\right) dk$$

$w = 0$: trivial $\quad$ $w = 1$: topological

Bulk–Boundary Correspondence

$w = 1 \Rightarrow$ one protected zero-energy state at each end of a finite chain. Protected by **chiral symmetry** $\{H, \Gamma\} = 0$.

Photonic Analogue

$$t_1 \leftrightarrow \frac{d_0}{a} = 0.45 \text{ (topological)}$$

$$t_2 \leftrightarrow \frac{d_0}{a} = 0.51 \text{ (trivial)}$$

2D SSH Model & Higher-Order Topological Insulator

2D SSH Hamiltonian (4-site unit cell)

Extend 1D SSH to a C_4-symmetric square lattice with:

- Intra-cell coupling: t_1
- Inter-cell coupling: t_2
- Next-nearest-neighbour: $t_{nn} = t_2/\sqrt{2}$

Topological phase: $|t_2| > |t_1|$

Nested Wilson Loops

Topological invariant computed via Wilson loop eigenvalues:

$$q_{xy} = \frac{1}{2} \pmod 1$$

Non-zero quadrupole moment q_{xy} signals higher-order topology, protected by C_4 and mirror symmetries.

Corner Charge (HOTI)

For C_4-symmetric HOTI, the bulk has **quantised corner charges**:

$$Q_{\text{corner}} = \frac{e}{4} \quad \text{(per corner)}$$

Four corners $\Rightarrow$ total charge $= e$. In photonics: H_z field localised at unit-cell corners.

NNN Coupling & Chiral Symmetry

From Supplementary Note 2 (Kivshar et al.): t_{nn} **breaks chiral symmetry**, causing in-gap disclination states to shift *within* the bulk gap. However:

- Filling anomaly persists
- Topological disclination states survive
- Energy is *tunable* by adjusting t_2/t_1

Disclination Defects: Fractional Charge & In-gap States

Volterra Process: $C_4 \to C_5$

Insert a Frank-angle sector $\Omega = \pi/2$:

1. Open a cut along $\theta = 0$
2. Insert angular sector $[0, \Omega]$ with C_4 tiling
3. Compress total angle $2\pi + \Omega \to 2\pi$ via scaling $\theta \to \theta \cdot \frac{2\pi}{2\pi+\Omega}$

Result: C_5-symmetric disclination with Frank angle $\Omega = +\pi/2$.

Bulk–Disclination Correspondence

Fractional charge at disclination core:

$$Q_{\mathrm{dis}} = \frac{p \cdot \Omega}{2\pi}$$

For C_5 ($\Omega = \pi/2$, $p = \frac{1}{2}$): $Q_{\mathrm{dis}} = \frac{e}{8}$. Guarantees **localised in-gap states** at the disclination core.

C_5 Angular Momentum States

C_5 symmetry $\Rightarrow$ 5 in-gap eigenstates:

l	Degeneracy	Mode type
0	1	Scalar mode
± 1	2	Vortex/anti-vortex
± 2	2	Higher-OAM

Energy Tunability (Supp. Fig. R1)

Even without chiral symmetry, in-gap states remain in the bulk gap. Their energy is tunable by:

- Adjusting t_2/t_1 ratio
- Modifying boundary hopping t'_w (air-hole size at disclination boundary)

This tunability is crucial for aligning the resonance with the THz bandgap.

Disclination States: Energy Tunability (Supplementary)

Broken Chiral Symmetry with NNN Coupling

The full Hamiltonian includes NNN hopping t_{nn}:

$$H = H_{\mathrm{SSH}} + t_{nn} \sum_{\langle\langle i,j \rangle\rangle} c_i^\dagger c_j$$

This breaks chiral symmetry, making the spectrum *asymmetric*. However, topological disclination states ($l = 0, \pm 1, \pm 2$) are **not eliminated** — they persist and are tunable within the gap.

Boundary Hopping Tuning

Modifying boundary hopping t'_w controls degeneracy splitting:

- Small $|t'_w|$: states degenerate, well-confined
- Large $|t'_w|$: states split, merge into bulk
- Experimentally: change air-hole radius at disclination boundary

Filling Anomaly

Even when in-gap modes shift toward bulk continuum, the topological invariant manifests as a **filling anomaly**: counting occupied states reveals fractional corner/disclination charge, independent of the gap position of in-gap modes.

Key Design Insight

The resonant frequency of the cavity is engineered to fall *within* the photonic bandgap of the bulk C_4 structure. This confinement mechanism — rather than total internal reflection alone — is what gives the high-Q factor. Adjusting:

$$\frac{r_0}{a}, \quad \frac{d_0}{a}, \quad t'_w$$

controls whether the resonance falls inside the gap.

Theory Summary

Step 1

2D SSH / HOTI

C_4 bulk lattice with quantised corner charge $Q = e/4$. Topological phase: $|t_2| > |t_1|$.

Step 2

Volterra Process

Insert $\Omega = \pi/2$ sector to obtain C_5 disclination. Bulk–disclination correspondence guarantees 5 in-gap states.

Step 3

Photonic Cavity

Replace lattice sites with air holes. Tune r_0/a, d_0/a so resonance falls in bandgap. Simulate with COMSOL 6.4.

Core Physical Principle

The **bulk** C_4 photonic crystal has a photonic bandgap. The **disclination defect** (after Volterra) hosts localised in-gap states due to fractional charge accumulation. Engineering the cavity geometry ensures these states fall *within* the bandgap, resulting in light confinement and high-Q resonances — without any mirrors or total internal reflection needed.